Dual Numbers

$\varepsilon^2 = 0$ $\qquad$ $\overset{0}{\downarrow}$

$$F(x+\varepsilon) = f(x) + f'(x)\varepsilon + \frac{f''(x)}{2}\varepsilon^2$$

$f: \mathbb{R}^n \to \mathbb{R}^m$ make a $\vec{x} + \vec{v}\varepsilon$

$\vec{v}$ can be $(1, \dots, 1)$ to get all or $(1, 0, \dots)$ to ge $\frac{\partial f}{\partial x_1}$, etc.

$x \in \mathbb{R}^n$ $\quad f(\vec{x} + \varepsilon\vec{v}) = f(\vec{x}) + Df(\vec{x})\vec{v}$

$$Df(x)v = J_f(x)\vec{v} \qquad J_f(x) \in \mathbb{R}^{m \times n}$$

$$(Jv)_j = \sum_{i=1}^{n} J_{ji} v_i \qquad (Df(x)v)_j = \sum_{i=1}^{n} \frac{\partial f_j}{\partial x_i} v_i$$

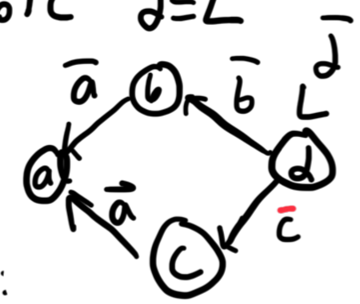
$\Theta(n)$, one for each basis e_i for dual

Back propagation (Reverse AD)

Nothing to do with dual #'s

"Forward": $b = a^2$ $\quad c = 3a$ $\quad d = b + c$ $\quad d = L$

think of this like chain rule



backwards: $= \frac{\partial L}{\partial u} = \bar{u} = \sum_n \frac{\partial L}{\partial v_n} \frac{dv_n}{du}$

$$\sum_n \bar{v}_n \frac{dv_n}{du}$$

a affects final output route through two different routes

Since in a program, you run like one at a time, you say $\bar{a} \mathrel{+}= \bar{b}\frac{db}{da}$ cuz you don't know if c if a fct of a

You get all of gradient $\Theta(m)$ time

$J \in \mathbb{R}^{n \times m}$ where in this case $m=1$ cuz scalar loss

Matrix Calculus

$$dL = \sum_{i,j} \frac{\partial L}{\partial X_{ij}} dX_{ij} = \mathrm{tr}((\nabla_X L)^T dX) \qquad x = \begin{bmatrix} x_1 \\ \vdots \end{bmatrix} \quad dx = \begin{bmatrix} dx_1 \\ \vdots \end{bmatrix}$$

$$L(x) = x^T x \qquad dL = (dx)^T x + x^T(dx)$$

$$\nabla_x L = 2x^T \qquad = 2x^T(dx)$$